# Introduction

The scalar curvature of a Riemannian Manifold $(M, g)$ is a smooth function $\operatorname{scal}\colon M \to \mathbb{R}$ measuring how much the volume of small geodesic balls around a point deviates from that of euclidian balls. While it is the weakest curvature condition for a Riemannian manifold, the presence of positive scalar curvature can still have strong topological implications. The archetypal example is the celebrated Gauß-Bonnet Theorem:

**Theorem** (Gauß-Bonnet). *Let $(M, g)$ be a closed Riemannian 2-manifold. Then*

$$\int_M \operatorname{scal} \mathrm{d}A = 4\pi\chi(M),$$

*where $\chi(M)$ is the Euler characteristic of $M$.*

For non-compact complete manifolds, merely having positive scalar curvature may be too weak of a condition. The problem is that scal can still tend to zero quite rapidly at infinity. However, if the rate of decay at infinity can be controlled, positive scalar curvature still has profound consequences. For example, Chen has recently shown a decomposition result for 3-manifolds under the assumption that a positive scalar curvature metric with controlled decay at infinity exists [Chen2025]. This naturally raises the question of how fast the scalar curvature of a given complete Riemannian manifold $X$ decays at infinity. One statement in this direction is the following theorem by Gromov:

**Theorem** (Enlargeable Decay Theorem, [**?**], page 15f.). *Let $X$ be a complete Riemannian manifold, $E \to M$ a two dimensional vector bundle over an enlargeable manifold $M$. Assume that*

*(i) There exists a proper map $f\colon X \to E$, $\deg(f) \neq 0$,*

*(ii) The total space of the circle bundle corresponding to $E$ is also enlargeable.*

*Then*

$$\min_{B_{x_0}(R)} \operatorname{scal}(X) \leq \frac{8\pi^2}{(R - R_0)^2}$$

*for some $R_0 = R_0(X, x_0)$ and all $R > R_0$.*

It is easy to see that condition (ii) is always fullfilled if the bundle $E \to M$ is trivial (compare Lemma 3.3 (i)), which lead Gromov to pose the following question:

**Question** ([**?**]). "What happens to nontrivial bundles $E \to M$? [...] In general, the examples in [**?**] indicate a possibility of non-enlargeable circle bundles over enlargeable $M$; yet, it seems hard(er) to find such examples, where the corresponding total space would admit metrics with scal $> 0$."

As it turns out, such examples do in fact exist, that is, circle bundles over enlargeable manifolds, where the total space admits a metric of positive scalar curvature and thus, in particular, is not enlargeable. Bundles with this behaviour have recently been constructed in a paper by Kumar and Sen [**?**] by utilizing tools from symplectic geometry. A very rough sketch of the construction can be given as follows. Take $M$ to be a so-called Donaldson divisor of a suitable closed symplectic manifold, e.g., the torus $T^{2n}$, $n \geq 3$. Such a Donaldson divisor is a certain codimension 2 symplectic submanifold $M^{2n-2} \subseteq T^{2n}$, which in some cases can be shown to be enlargeable. Now define a circle bundle $E := \partial\nu(M)$ over $M$ as the boundary of a tubular neighbourhood $\nu(M) \subseteq T^{2n}$ of $M$. In order to construct a positive scalar curvature metric on $E$, consider the nullbordism $W := T^{2n} - \nu(M)$ obtained by removing the tubular neighbourhood from the ambient manifold. Now explicitly construct a Morse function $\psi\colon W \to \mathbb{R}$ with critical points of index at most $n$, which shows that $E$ is obtained from the empty set by surgeries of codimension at least $n \geq 3$. Thus, $E$ admits a metric of positive scalar curvature by the Gromov-Lawson-Schoen-Yau Surgery Theorem (see Theorem 2.1).

In this thesis, we provide an alternative, much more direct construction relying simply on basic topological concepts. It should be said, however, that this alternative approach only provides examples in the totally non-spin case.

The thesis is organized as follows. Chapter 1 recalls some concepts from the broader theory of principal bundles, laying the foundation for the subsequent chapters. In Chapter 2, we briefly introduce the reader to the theory of positive scalar curvature and prove two existence theorems for positive scalar curvature metrics, which are used in the main construction in Chapter 4. The proofs of these theorems are based on methods from surgery theory, which we also take some time to explain. Chapter 3 provides a brief discussion of the concept of enlargeability, which is the central idea underlying this thesis. Finally, Chapter 4 contains the main results. We present a simple construction of circle bundles over enlargeable manifolds whose total spaces admit metrics of positive scalar curvature in all dimensions $\geq 4$. These bundles are then used to obtain counterexamples to the Enlargeable Decay Theorem above if condition (ii) is dropped.

# 1 Classifying Spaces and Characteristic Classes

This section will lay the groundwork for the subsequent material, by recalling the notions of principal bundles, classifiying spaces and characteristic classes. We assume the reader to be familiar with the concept of a fiber bundle with structure group. A classical reference for this is for example [?].

## 1.1 Principal Bundles and Associated Bundles

**Definition 1.1.** Let $G$ be a topological group. A $G$-principal bundle is a fiber bundle with fiber $G$ and structure group $G$ acting by left translations.

The total space of a $G$-principal bundle $P \xrightarrow{\pi} B$ carries a natural right action of $G$ defined as follows. Around $y \in P$ choose a local trivialisation

$$\phi\colon U \times G \to \pi^{-1}(U)$$

and let $\phi^{-1}(y) = (\pi(y), h)$. Then set

$$y \cdot g := \phi(\pi(y), hg).$$

This is well defned since by definition $G$ acts on itself by *left* translations under a change of local trivialisation. A morphism of principal bundles should respect this right action, thus we define:

**Definition 1.2.** A morphism from a $G$-principal bundle $P \to B$ to a $G$-principal bundle $P' \to B'$ is a pair $(F, f)$ of continuous maps $F\colon P \to P'$ and $f\colon B \to B'$, such that

$$\begin{array}{ccc} P & \xrightarrow{F} & P' \\ \downarrow & & \downarrow \\ B & \xrightarrow{f} & B' \end{array}$$

commutes and such that $F$ is equivariant, that is $F(x \cdot g) = F(x) \cdot g$ for all $x \in P$ and $g \in G$. If $B = B'$ we assume $f$ to be the identity.

The condition of equivariance in the above definition is actually quite strong as illustrated by the following proposition.

**Proposition 1.3.** *Any morphism of $G$-principal bundles over the same base space is an isomorphism.*

*Proof.* First suppose we are given a morphism

$$F\colon B \times G \to B \times G$$

of trivial bundles. Then by definition $F$ is of the form

$$F(x, g) = (x, \phi_x(g))$$

for a continuous family of equivariant maps $\phi_x\colon G \to G$. By equivariance $\phi_x(g) = \phi_x(e)g$ and thus an inverse of $F$ is given by

$$F^{-1}(x, g) := (x, \phi_x(e)^{-1}g).$$

Since every $G$-principal bundle is locally trivial this shows that we can always find local inverses in the general case and since inverses are unique they fit together to define a global inverse. □

ÜBERLEITUNG!! Let $p\colon E \to B$ be a fiber bundle with fiber $F$ and structure group $G$ and let $F'$ be another space upon which $G$ acts effectively. Then we can construct a new bundle as follows. Choose an open cover $(U_\alpha)_{\alpha \in A}$ of $B$ together with local trivialisations $\phi_\alpha\colon U_\alpha \times F \to p^{-1}(U_\alpha)$. Then there are transition functions $\tau_{\alpha\beta}\colon U_\alpha \cap U_\beta \to G$ defined by

$$(\phi_\beta^{-1} \circ \phi_\alpha)(x, f) = (x, \tau_{\alpha\beta}(x) \cdot f).$$

Define

$$E' := \left( \coprod_{\alpha \in A} U_\alpha \times F' \right) \Big/ \sim$$

where the equivalence relation $\sim$ is defined by

$$(b, f', \beta) \sim (b, \tau_{\alpha,\beta}(b) \cdot f', \alpha)$$

and let $p' \colon E' \to B$, $[(b, f', \alpha)] \mapsto b$. It is then not too hard to show, that $p' \colon E' \to B$ can be given the structure of a fiber bundle with fiber $F'$ and structure group $G$ with the same transition functions $\tau_{\alpha\beta}$ (QUELLE). We say that this new bundle is obtained from $E \to B$ by *changing the fiber* from $F$ to $F'$ (with the $G$-action on $F'$ be implicit). Note that we can obtain $E \to B$ back from $E' \to B$ by changing the fiber from $F'$ back to $F$. Thus no information is lost in this process. Considering the special case $F' = G$ with $G$ acting on itself by left translations, we call the bundle obtained from $E \to B$ by changing the fiber from $F$ to $G$ the *associated principal bundle*. Morally speaking if we want to understand the bundle $E \to B$ it is sufficient to understand its associated prinicpal bundle, as it contains the same information as we have seen above. The benefit of studying principal bundles is that they are very symmetric obejcts, making them easier to understnad. For example many constructions simlify when performed on principal bundles, such as the ... of universal bundles as we will see in the next subsection. Another such instance is the construction of changing the fiber from above, which can be made more dirctly in the case of principal bundles. This is known as the *Borel Construction*:

**Lemma 1.4** (Borel Construction, QUELLE?)**.** *Let $P \xrightarrow{p} B$ be a $G$-principal bundle and $F$ a space upon which $G$ acts effectively from the left. Then*

$$E := (P \times F)\big/_{\sim}, \quad (y, f) \sim (y \cdot g, g^{-1} f)$$

*together with the map $\pi \colon E \to B$ given by $[(y, f)] \mapsto p(y)$ is a fiber bundle with fiber $F$ and structure group $G$. Moreover this bundle is isomorphic to the bundle obtained from $P \to B$ by changing the fiber from $G$ to $F$ using the construction outlined above.*

The Borel construction is the reason why the bundle obtained from $P \to B$ by changing the fiber from $G$ to $F$ is usually denoted by $P \times_G F \to B$. With this notation in mind we can formulate an important naturality property regarding the change of fibers.

**Lemma 1.5.** *Let $P \to B$ be a $G$-prinicpal bundle, $F$ a space upon which $G$ acts effectively and $f \colon B' \to B$ a continuous map. Then*

$$(f^* P) \times_G F \cong f^*(P \times_G F).$$

*Proof.* (Using the explicit despriction of $P \times_G F$ from Lemma 1.4 this should be relatively straightforward) $\square$

Lets look at some concrete examples for the theory developed above.

**Example 1.6.** For simplicity we assume all base spaces $B$ to be smooth manifolds in this example.

i) Let $E \to B$ be a real rank $n$ vector bundle. Choosing a bundle metric gives $E \to B$ the structure of a fiber bundle with fiber $\mathbb{R}^n$ and structure group $O(n)$. The associated principal bundle is explicitly given by the *orthonormal frame bundle of* $E$, whose fiber over a point $x \in B$ is the space of all ordered bases of $E_x$.

ii) An orientation of a real vector bundle allows us to further reduce the structure group to $SO(n)$. The associated principal bundle is then given by the *oriented orthonormal frame bundle of* $E$, whose fiber over a point $x \in B$ is the space of all oriented ordered bases of $E_x$.

iii) An example, which will be of relevance later is the following. Let $P \to B$ be an $S^1$-prinicpal bundle. We can change the fiber of $P$ from $S^1$ to $D^2$ to obtain another bundle $E \to B$, where $S^1$ acts on $D^2$ by rotations. Since this action just extends the action of $S^1 = \partial D^2$ on itself we see that $P \cong \partial E$. So in particular the total space of any $S^1$-prinicpal bundle is null-bordant.

iv) The associated prinicpal bundle of the tautological line bundle $\xi_1^n \to \mathbb{C}P^n$ is given by the projection map $S^{2n+1} \to \mathbb{C}P^n$ dividing out the $S^1$ action on $S^{2n+1}$. PROOF

## 1.2 Universal Bundles and Classifying Spaces

A natural question is how many bundles exist... again easier for principal bundles... For simplicity we assume all base spaces to be CW-complexes in this section.

**Definition 1.7.** A $G$-principal bundle $E \to B$ is called universal if $E$ is weakly contractible as a topological space.

The significance of this definition is established by the following theorem. We say that two morphisms of $G$-principal bundles are $G$-homotopic if they are homotopic through morphisms of $G$-principal bundles.

**Theorem 1.8.** *Let $E \to B$ be a universal $G$-principal bundle. Then for any other $G$-principal bundle $P \to X$ there exists a morphism $P \to E$ of $G$-principal bundles, which is unique up to $G$-homotopy.*

Before showing the Theorem let us discuss its implications. Firstly we can equivalently formulate the Theorem by saying that $E \to B$ is a terminal object in the homotopy category of $G$-principal bundles over CW-complexes and thus in particular uniquely determined (up to $G$-homotopy equivalence). This justifies writing $EG := E$ and $BG := B$. Since any bundle morphism $(F, f)\colon P \to EG$ induces a morphism $P \to f^*EG$ (and vice versa, which is true for arbitrary fiber bundles) we moreover conclude from Proposition 1.3 that

$$P \cong f^*EG.$$

We call $f$ a **classifiying map** for $P$. By homotopy invariance of the pullback we get a well-defined bijection

$$\phi_X\colon [X, BG] \to \mathcal{P}_G(X),\ [f] \mapsto [f^*EG],$$

where $\mathcal{P}_G(X)$ denotes the set of isomorphism classes of $G$-principal bundles over $X$. To see that $\phi_X$ is indeed bijective note that the existence result in Theorem **??** shows that $\phi_X$ is surjective and the uniqueness result shows that it is injective. Moreover $\phi_X$ is clearly natural in $X$. Thus we have shown.

**Corollary 1.9.** *The maps $\phi_X$ define a natural isomorphism of contravaraint functors*

$$[-, BG], \mathcal{P}_G(-)\colon \mathsf{HCW} \to \mathsf{Set},$$

*where* $\mathsf{HCW}$ *denotes the homotopy category of CW-complexes.*

Theorem 1.8 will be an application of the following Theorem.

**Theorem 1.10.** *Let $P \to X$ be a $G$-principal bundle and $A \subseteq X$ a subcomplex. Then any morphism $P|_A \to EG$ can be extended to a morphism $P \to EG$.*

A proof of this Theorem based on an inductive cell by cell approach can be found in [**?**, Theorem 2.7.1].

*Proof of Theorem 1.8.* Applying Theorem 1.10 to $A = \emptyset$ shows the existence of a morphism $P \to EG$. Now suppose we are given morphisms $F_0, F_1\colon P \to EG$. Consider the $G$-principal bundle

$$\pi \times \mathrm{id}\colon P \times [0,1] \to X \times [0,1],$$

where $\pi\colon P \to X$ is the bundle projection (this is just the pullback bundle of $P$ under the projection $X \times [0,1] \to X$). Then $F_0$ and $F_1$ define a morphism of $G$-principal bundles

$$(F_0, F_1)\colon P \times \{0,1\} = (P \times [0,1])|_{X \times \{0,1\}} \to EG,$$

which by Theorem 1.10 can be extended to $P \times [0,1]$, thus yielding a $G$-homotopy between $F_0$ and $F_1$. □

of course for this theoy to be of any use we need such universal bundles to exist...

**Theorem 1.11** (Milnor construction)**.** *For any topological group $G$ there exists a universal $G$-principal bundle $EG \to BG$, where $BG$ is a CW-complex.*

understood principal bundles, how can we recover information abotu general bundles –¿ changing the fiber

**Corollary 1.12.** *Let $G$ be a topological group acting effectively on a space $F$. Then for any CW-complex $X$ the map*

$$[X, BG] \to \mathcal{F}_{F,G}(X), \;\; [f] \mapsto [f^*(EG \times_G F)]$$

*is a natural bijection, where $\mathcal{F}_{F,G}(X)$ denotes the set of isomorphism classes of fiber bundles with fiber $F$ and structure group $G$ over $X$.*

*Proof.* By 1.9 the map

$$[X, BG] \to \mathcal{P}_G(X), \;\; [f] \mapsto [f^*EG]$$

is a natural bijection and by Lemma 1.5

$$\mathcal{P}_G(X) \to \mathcal{F}_{F,G}(X), \;\; [P] \mapsto [P \times_G F]$$

is a natural bijection. Composing both thus yields the natural bijection

$$[X, BG] \to \mathcal{F}_{F,G}(X), \;\; [f] \mapsto [(f^*EG) \times_G F]\,.$$

So the statement follows by again applying Lemma 1.5. □

Let us look at some examples...

**Example 1.13.** label

*) $BO(n)$ and $EO(n)$ explicit via Stiefel Manifolds (compare n lab) + $EO(n) \times_{O(n)} \mathbb{R}^n \cong \xi_n^\infty$ for $\xi_n^\infty$ tautological bundle over grassmanain

*) Analogous for complex case??

*) Special cases $\mathbb{R}P^\infty$ and $\mathbb{C}P^\infty$

*) $M \to B\pi_1(M)$ induces iso on $\pi_1$

## 1.3 Stiefel-Whitney and Chern-Classes

We have seen how Another fundamental question is of course how to decide, whether two bundles are isomorphic. As usual in topology, one tries to define certain invariants posing as an obstruction... For vector bundles (or more generally prinicpal bundles), ... characteristic classes. The general definiton of a characteritsic class is as follows. Let $\mathbb{K} = \mathbb{R}, \mathbb{C}$ and write $\mathrm{Vect}_{\mathbb{K}}(B)$ for the set of isomorphism classes of $\mathbb{K}$-vector bundles over a space $X$. Then $\mathrm{Vect}_{\mathbb{K}}(-)$ defines a contravariant functor $\mathsf{Top} \to \mathsf{Set}$. A characteristic class $c$ for $\mathbb{K}$-vector bundles with values in the cohomology theory $H^*$ is a natural transformation from the functor $\mathrm{Vect}_{\mathbb{K}}(-)$ to the functor $H^*$ regarded as a functor with values in $\mathsf{Set}$. So in more concrete terms $c$ assigns to each $\mathbb{K}$-vector bundle $E \to B$ a cohomology class $c(E) \in H^*(B)$ depending only on the isomorphism class of $E$, such that for any continuous map $f\colon B' \to B$ the following naturality condition holds

$$c(f^*E) = f^*c(E).$$

So in particular, using the theory of classifiying spaces developed above, we see that for a real (resp. complex) rank $k$ vector bundle $E \to B$, the class $c(E)$ is uniquely determined by the value $c(\xi_k^\infty) \in H^*(BO(k))$ for the universal bundle $\xi_k^\infty$ introduced in example...

For real vector bundles the most important characteristic classes are the so called *Stiefel-Whitney classes*, which can be completely characterized by a few axioms.

**Definition 1.14** (Stiefel-Whitney Classes, HATCHER)**.** There is a unique sequence of characteristic classes $w_1, w_2, \ldots$ for real vector bundles with values in $H^*(-;\mathbb{Z}_2)$, such that

i) $w_i(E) \in H^i(B;\mathbb{Z}_2)$ for any real bundle $E \to B$,

ii) $w_i(E) = 0$ for $i > \mathrm{rank}(E)$,

iii) $w(E_1 \oplus E_2) = w(E_1) \smile w(E_2)$, where $w = 1 + w_1 + w_2 + \ldots$,

iv) For the canonical line bundle $\xi_1^\infty \to \mathbb{R}P^\infty$, $w_1(E)$ is the generator of $H^1(\mathbb{R}P^\infty; \mathbb{Z}_2) \cong \mathbb{Z}_2$.

Note that by property ii) the sum $w = 1 + w_1 + w_2 + \ldots$ is always finite. This class $w(E)$ is called the *total Stiefel-Whitney class of E*. For obvious reasons one sometimes writes $w_0(E) := 1 \in H^0(B; \mathbb{Z}_2)$. The analogues for complex vector bundles are known as *Chern classes.*

**Definition 1.15** (Chern Classes, HATCHER)**.** There is a unique sequence of characteristic classes $c_1, c_2, \ldots$ for complex vector bundles with values in $H^*(-; \mathbb{Z})$, such that

i) $c_i(E) \in H^{2i}(B; \mathbb{Z})$ for any complex bundle $E \to B$,

ii) $c_i(E) = 0$ for $i > \mathrm{rank}(E)$,

iii) $c(E_1 \oplus E_2) = c(E_1) \smile c(E_2)$, where $c = 1 + c_1 + c_2 + \ldots$,

iv) For the canonical line bundle $\tau_1^\infty \to \mathbb{C}P^\infty$, $c_1(E)$ is a generator of $H^2(\mathbb{C}P^\infty; \mathbb{Z}) \cong \mathbb{Z}$ specified in advance.

Again $c = 1+c_1+c_2+\ldots$ is well-defined and is called the *total Chern class.* Again one sometimes writes $c_0(E) := 1 \in H^0(B; \mathbb{Z})$. See [HATCHER] for a proof of existence and uniqueness.

**Example 1.16.**

i) Let $\iota\colon \mathbb{R}P^n \to \mathbb{R}P^\infty$ be the canonical inclusion. Then the pullback $\iota^*\xi_1^\infty \cong \xi_1^n$ is the tautological line bundle $\xi_1^n \to \mathbb{R}P^n$. Since $\iota$ induces an isomorphism on cohomology we conclude from naturality that $w_1(\xi_1^n) = \iota^* w_1(\xi_1^\infty)$ is the generator of $H^n(\mathbb{R}P^n; \mathbb{Z}_2) \cong \mathbb{Z}_2$. Similiarly $c_1(\tau_1^n)$ of the tautological complex line bundle $\tau_1^n \to \mathbb{C}P^n$ is a generator of $H^2(\mathbb{C}P^n; \mathbb{Z}) \cong \mathbb{Z}$.

ii) Let $\varepsilon \to B$ be a trivial real bundle over some space $B$. Then $\varepsilon \cong f^*\varepsilon$ for a constant map $f\colon B \to B$ and thus $w_i(\varepsilon_n) = f^* w_i(\varepsilon) = 0$ for all $i \geq 1$. Similiarly all Chern classes for complex trivial bundles vanish.

iii) Let $TS^n \to S^n$ be the tangent bundle of the sphere and let $\nu_n \to S^n$ be the normal bundle. Since $\nu_n$ and $TS^n \oplus \nu_n$ are trivial we conclude from the example above and property iii) in Definition 1.14 that

$$1 = w(TS^n \oplus \nu_n) = w(TS^n) \smile w(\nu_n) = w(TS^n) \smile 1 = w(TS^n).$$

Thus $w_i(TS^n) = 0$ for all $i \geq 1$.

In the last example we have seen that although all Stiefel-Whitney classes of a bundle vanish, the bundle need not be trivial. However for one-dimensional real (resp. complex) bundles it turns out that they are already completely classified by their first Stiefel-Whitney class (resp. Chern class):

**Theorem 1.17** (Hathcer, Proposition 3.10)**.** *Let X be a CW-complex. Then the maps*

$$w_1\colon \mathrm{Vect}_{\mathbb{R}}(X) \to H^1(X; \mathbb{Z}_2)$$

*and*

$$c_1\colon \mathrm{Vect}_{\mathbb{C}}(X) \to H^2(X; \mathbb{Z})$$

*are bijections.*

This allows us to define line bundles by only prescribing their first Stiefel-Whitney class (resp. Chern class), which is often technically easier thank directly writing down this bundle. While the first Stiefel-Whitney class of higher dimensional bundles does not identify their isomorphism type it does characterize another important invariant, namely orinetability:

**Proposition 1.18** (Hatcher, Prop 3.11)**.** *Let X be a CW-complex. A real vector bundle $E \to X$ is orientable if and only if $w_1(E) = 0$.*

Similiarly vanishing of the second Stiefel Whitney class $w_2(E)$ corresponds to the existence of another structure on $E$, namely a spin structure. We need not be concerned with what precisely a spin structure is, such that the following definition suffices for our pruposes.

**Definition 1.19.** An orientable manifold $M$ is called

i) *spin*, if $w_2(TM) = 0$,

ii) *totally non-spin*, if the universal cover of $M$ is not spin.

Recall that a manifold $M$ is orientable if and only if its tangent bundle $TM$ restricted to any embedded loop $S^1 \subseteq M$ is trivial. Or in other words "$TM$ does not contain any Moebius Bands". A similiar characterization can be given for spin manifolds, which is why the property of being spin is sometimes referred to as a higher dimensional analogue of orientability.

**Proposition 1.20** (Georg Paper, Proposition 2.6)**.** *Let $M$ be a connected oriented manifold of dimension at least $5$.*

*i) $M$ is spin if and only if for every embedded surface $S \subseteq M$ the restriction $TM|_S$ of the tangent bundle to $S$ is trivial.*

*ii) $M$ is totally non-spin if and only if there exists an embedded sphere $S^2 \subseteq M$, such that the restriction $TM|_{S^2}$ of the tangent bundle to this sphere is trivial.*

*Proof.* Since we will not be making use of the first characterization of spin manifolds, we shall only proof part ii) of the proposition and refer the reader to [Georg Paper] for a proof of part i). ...(Proof) □

Finally we record another scenario in which the vanishing of all Stiefel-Whitney classes of a bundle is sufficient for triviality.

**Lemma 1.21.** *Let $E \to S$ be a real vector bundle over a surface $S$ of rank at least $3$. Then $E$ is trivial if and only if $w_1(E) = w_2(E) = 0$.*

proof...

# 2 Positive Scalar Curvature

## 2.1 Overview

brief definition, maybe intuition and osbtructions/backgroudn etc. based on overview paper

psc is somehow sweetspot for rigidity and flexibility

Probably the most important existence Theorem for positive scalar curvature metrics... Before stating the theorem let us briefly recall what surgery theory is about. The basic oberervation ist that a product $S^k \times S^{n-k-1}$ of spheres can be considered as the boundary of $S^k \times D^{n-k}$ or of $D^{k+1} \times S^{n-k-1}$... davor noch sowas schreiben wie "change the topology of a manifold in a controlled way"; remove thickened embedded $k$-sphere ($\rightarrow$ terminology of surgery of dimension k); EMBEDDINGS ALWAYS INTO INTERIOR (dann muss ich das unten in proposition zur surgery nicht mehr immer dazu schreiben) + can extend embedding of sphere iff trivial normal bundle; surgery intinmately connected to cobordism theory; how does surgery correspond to handles?; one common application of surgery theory is to "kill" certain homotopy groups $\rightarrow$ explain this process more precisely below

**Theorem 2.1** (Schoen,Yau 1979, Gromov,Lawson,1980)**.** *(Surgery Theorem)*

In order to apply this theorem we need a way to control the dimension of the surgeries performed or equivalently the indices of handles in a given cobordism. This question is completely independent of scalar curvature and has been subject of research especially in the context of the $h$-cobordims theorem. One central tool to control the indices of the handles in a cobordism is the following theorem due to Wall

**Theorem 2.2** (Wall, 1971)**.** *(Wall Theorem)*

Combining gromov,lawson and Wall we see that: $M_0$ is psc, $M_1 \rightarrow W$ 2-connected then $M_1$ psc. In fact one can use the Theorem of Wall to obtain an even stronger result:

**Theorem 2.3** (extending psc to corbodism, Georg,Gaja)**.** ...

So in view of the above discussion, the strategy to construct a psc metric on a given closed manifold $M_1$ of dimension at least 5 is to first find a cobordism $W$ from some psc manifold $M_0$ to $M_1$ and then modify this cobordism by surgeries to obtain another cobordism $W'$ from $M_0$ to $M_1$, such that the inlcusion $M_1 \rightarrow W'$ is 2-connected. Of course, such a modification by surgeries may not always be possible, so our next goal is to explain certain scenarios in which it is. In order to formulate these we recall the following definition.

**Definition 2.4** (Spin and Totally non spin)**.** ...

We now have the following two existence theorems for psc metrics. One for the spin case and one for the totally non-spin case.

**Theorem 2.5.** *Let $W^{n+1}$ be a cobordism from $M_0$ to $M_1$, $n \geq 5$, such that $W$ is spin. Suppose that $M_1$ is connected, the inlcusion $M_1 \rightarrow W$ induces an isomorphism on $\pi_1$ and that $M_0$ carries a psc-metric. Then $M_1$ also carries a psc-metric. In fact the psc-metric on $M_0$ can be extended to a psc-metric on $W$, which is of product form near the boundary.*

**Theorem 2.6.** *Let $W^{n+1}$ be an oriented cobordism from $M_0$ to $M_1$, $n \geq 5$. Suppose that $M_1$ is a connected totally non-spin manifold, the inlcusion $M_1 \rightarrow W$ induces an isomorphism on $\pi_1$ and that $M_0$ carries a psc-metric. Then $M_1$ also carries a psc-metric. In fact the psc-metric on $M_0$ can be extended to a psc-metric on $W$, which is of product form near the boundary.*

By the general strategy we laid out above, the proofs of these theorems will be using the given data to perform surgery on the cobordism $W$, such that the inlcusion $\iota\colon M_1 \rightarrow W$ becomes 2-connected. By assumption $M_1$ is connected and $\iota$ already induces an isomorphism on $\pi_1$, so formulated differently our goal is to "kill" all generators in $\pi_2(W, M)$ by surgery. So as a first step let us make precise how one can use surgery to "kill" certain homotopy classes.

**Proposition 2.7.** *Let $W^n$ be smooth manifold with basepoint $w_0 \in W$, $M \subseteq \partial W$ a boundary component with basepoint $x_0 \in M$ and $2(k+1) \leq n$.*

*i) Suppose we are given classes $\alpha_1, \dots, \alpha_l \in \pi_k(W, w_0)$, which are represented by embeddings with trivial normal bundle. Then there exists a manifold $W'$ obtained from $W$ by $l$-many $k$-surgeries away from the point $w_0$ together with an epimorphism*

$$\pi_k(W, w_0) \twoheadrightarrow \pi_k(W', w_0)$$

*containing the $\mathbb{Z}[\pi_1(W, w_0)]$-module generated by $\alpha_1, \dots, \alpha_l$ in its kernel. Moreover*

$$\pi_i(W, w_0) \cong \pi_i(W', w_0)$$

*for all $i \leq k-1$.*

*ii) Suppose we are given classes $\alpha_1^{\mathrm{rel}}, \dots, \alpha_l^{\mathrm{rel}} \in \pi_k(W, M, x_0)$, which lift to classes $\alpha_1, \dots, \alpha_l \in \pi_k(W, x_0)$ under the map $\pi_k(W, x_0) \to \pi_k(W, M, x_0)$ induced by the inclusion. Again assume that $\alpha_1, \dots, \alpha_l$ are represented by embeddings with trivial normal bundle. Then there exists a manifold $W'$ obtained from $W$ by $l$-many $k$-surgeries together with an epimorphism*

$$\pi_k(W, M, x_0) \twoheadrightarrow \pi_k(W', M, x_0)$$

*containing the $\mathbb{Z}[\pi_1(M, x_0)]$-module generated by $\alpha_1^{\mathrm{rel}}, \dots, \alpha_l^{\mathrm{rel}}$ in its kernel. Moreover*

$$\pi_i(W, M, x_0) \cong \pi_i(W', M, x_0)$$

*for all $i \leq k-1$.*

Proof of proposition 2.7

Proof of Theorem 2.6 and 2.5

In the main construction in section ... we will invoke Theorem 2.6 to construct a psc metric, so let us conclude this section by discussing some basic results for identifying totally non-spin manifolds.

**Lemma 2.8.** *(i) The product of a totally non-spin manifold with any non-empty manifold is totally non-spin.*

*(ii) A manifold containing a totally non-spin submanifold of codimension* 0 *is totally non-spin.*

*(iii) If $M$ is totally non-spin and of dimension $n \geq 3$, then $M - D^n$ is totally non-spin for any embedded disk $D^n \subseteq \mathrm{int}(M)$ in the interior of $M$.*

*Remark* 2.9. All statements of the above lemma remain valid if “totally non-spin” is replaced by “non-spin”. The corresponding proofs are just simpler versions of the arguments given below.

*Proof.* For (i) let $M_1$ and $M_2$ be manifolds with $M_1$ totally non-spin. Then the universal cover of the product $M_1 \times M_2$ is given by the product $\tilde{M}_1 \times \tilde{M}_2$ of the respective universal covers. Denote by

$$\pi_1 \colon \tilde{M}_1 \times \tilde{M}_2 \to \tilde{M}_1, \quad \pi_2 \colon \tilde{M}_1 \times \tilde{M}_2 \to \tilde{M}_2$$

the projections and let $\iota \colon \tilde{M}_1 \to \tilde{M}_1 \times \tilde{M}_2$ be an inclusion for some choice of basepoint in $\tilde{M}_2$. We compute

$$\begin{aligned} \iota^* w_2(T(\tilde{M}_1 \times \tilde{M}_2)) &= \iota^* w_2(\pi_1^* T\tilde{M}_1 \oplus \pi_2^* T\tilde{M}_2) \\ &= w_2((\pi_1 \circ \iota)^* T\tilde{M}_1 \oplus (\pi_2 \circ \iota)^* T\tilde{M}_2) \\ &= w_2(T\tilde{M}_1) \neq 0, \end{aligned}$$

where we used that $\pi_1 \circ \iota$ is the identity and that $\pi_2 \circ \iota$ is constant, such that $(\pi_2 \circ \iota)^* T\tilde{M}_2$ is trivial. Thus $w_2(T(\tilde{M}_1 \times \tilde{M}_2)) \neq 0$.

For (ii) suppose that $N \subseteq M$ is some totally non-spin codimension 0 submanifold of the manifold $M$. Let $\iota \colon N \to M$ be the inclusion and consider its lift to the universal covers:

$$\begin{array}{ccc} \tilde{N} & \xrightarrow{\tilde{\iota}} & \tilde{M} \\ \downarrow & & \downarrow \\ N & \xrightarrow{\iota} & M \end{array}$$

Since $\tilde{\iota}$ is a local diffeomorphism, the differential $d\tilde{\iota}\colon T\tilde{N} \to T\tilde{M}$ is a fiber-wise isomorphism, such that the following is a pullback diagram

$$\begin{tikzcd} T\tilde{N} \arrow[r, "d\tilde{\iota}"] \arrow[d] & T\tilde{M} \arrow[d] \\ \tilde{N} \arrow[r, "\tilde{\iota}"] & \tilde{M} \end{tikzcd}$$

Therefore $\tilde{\iota}^* w_2(T\tilde{M}) = w_2(T\tilde{N}) \neq 0$ and so $w_2(T\tilde{M}) \neq 0$.

For (iii) assume without of loss generality that $M$ is connected. Denote by $B^n := \mathrm{int}(D^n)$ the open ball. Then $M - D^n$ is the interior of $M - B^n$ from which we conclude that $M - D^n$ is totally non-spin if and only if $M - B^n$ is totally non-spin. This follows using an analogous argument as in (ii) by noting that for a manifold $X$ with boundary the universal cover of $\mathrm{int}(X)$ is given by $\mathrm{int}(\tilde{X})$ and the inlcusion $\mathrm{int}(\tilde{X}) \to \tilde{X}$ is a homotopy equivalence. We will now show that $M - B^n$ is totally non-spin. Let $p\colon \tilde{M} \to M$ be the universal cover of $M$ and write $M_0 := M - B^n$. We claim that the universal cover of $M_0$ agrees with the covering

$$\hat{M}_0 = \tilde{M}|_{M_0} = \tilde{M} - p^{-1}(B^n) \xrightarrow{p} M_0.$$

First of all $\hat{M}_0$ is connected as $n \geq 2$. We now have to show that $\hat{M}_0$ is simply connected. Let $F = p^{-1}(x)$ be some fiber of $p$. From the classification of coverings we see that

$$p^{-1}(B^n) \cong B^n \times F$$

since $B^n$ is simply connected. If the fiber $F$ is finite we can apply the Seifert-van-Kampen Theorem to conclude that $\tilde{M}$ with a single disk $B^n$ removed is still simply connected, such that the statement follows inductively. If the fiber is not finite we can argue more generally as follows. Let $\phi\colon S^1 \to \hat{M}_0$ be some smooth loop. Certainly we can contract this loop in $\tilde{M}$, that is we can find some map $\Phi\colon D^2 \to \tilde{M}$ restricting to $\alpha$ on the boundary. We now show that $\Phi$ is homotopic $\mathrm{rel}\, S^1$ to some map $\Phi_1\colon D^2 \to \tilde{M}$, such that the image of $\Phi_1$ is already contained in $\hat{M}_0$, from which the claim follows. To construct such a homotopy let $0 \in B^n \subseteq M$ be the center of the embedded disk. After a preliminary homotopy we may assume that $\Phi$ is transversal to $p^{-1}(0)$. The crucial observation is that, since $n \geq 3$, this is equivalent to requiring that $\Phi$ does not intersect $p^{-1}(0)$. Now note that $S^{n-1}$ is a strong deformation retract of $D^n - 0$ and applying such a deformation retraction on each disk $D^n - 0 \subseteq p^{-1}(D^n - 0)$ shows that $\hat{M}_0$ is a strong deformation retract of $\tilde{M} - p^{-1}(0)$. Let

$$r\colon \tilde{M} - p^{-1}(0) \to \hat{M}_0$$

be such a retraction and let

$$r_t\colon \tilde{M} - p^{-1}(0) \to \tilde{M} - p^{-1}(0)$$

be a homotopy $\mathrm{rel}\, S^1$ from the identity to $r$ (composed with the corresponding inclusion). Then $\Phi_t := r_t \circ \Phi$ defines a homotopy $\mathrm{rel}\, S^1$ from $\Phi$ to some map $\Phi_1$ whose image is contained in $\hat{M}_0$. This shows the claim and hence the universal cover of $M_0$ is given by

$$\hat{M}_0 = \tilde{M} - p^{-1}(B^n) \cong \tilde{M} - (B^n \times F).$$

Finally, the Mayer-Vietoris sequence of the pushout

$$\begin{tikzcd} S^{n-1} \times F \arrow[r] \arrow[d] & D^n \times F \arrow[d] \\ \tilde{M} - (B^n \times F) \arrow[r, "\iota"] & \tilde{M} \end{tikzcd}$$

where all maps are inclusions, then shows that $\iota$ is injective on $H^2(-;\mathbb{Z}_2)$. Therefore

$$w_2(T\hat{M}_0) = w_2(\iota^* T\tilde{M}) = \iota^* w_2(T\tilde{M}) \neq 0.$$

$\square$

A standard example of a non-spin manifold is $\mathbb{C}P^2$. Since $\mathbb{C}P^2$ is simply connected this is also an example of a totally non-spin manifold, which we will appeal to in section ...

**Lemma 2.10.** *$\mathbb{C}P^2$ is non-spin.*

(proof)

# 3 Enlargeability

In this section we discuss the concept of enlargeability, which is the central idea the results in this thesis are concerned with.

**Definition 3.1.** A closed oriented Riemannian $n$-manifold $M$ is called enlargeable if for all $\varepsilon > 0$ there exists a Riemannian covering $\hat{M} \to M$ together with an $\varepsilon$-contracting (that is $\varepsilon$-Lipschitz) map $\hat{M} \to S^n$, which is constant outside a compact subset and of non-zero degree.

Note that this notion does not depend on the particular choice of Riemannian metric on $M$, since for closed manifolds all metrics are in bi-Lipschitz correspondence. Intuitively, the covering $\hat{M}$ can be pictured as an "expansion" of $M$ itself (think for example of the usual sketches of coverings of $S^1$). Enlargeability now means that this expansion can be chosen arbitrarily large, such that it can be wrapped as tightly around a sphere as one wishes. Naively one might expect that for an enlargeable manifold $M$ the universal cover $\tilde{M}$ - which in some sense is the "largest" covering - is hyperspherical, meaning that it admits an $\varepsilon$-contracting map $\tilde{M} \to S^n$, which is constant outside a compact subset and of non-zero degree for arbitrarily small $\varepsilon > 0$. However, Brunnbauer and Hanke have constructed counterexamples to this naive expectation in [BH, Theorem 1.4].

The simplest example of an enlargeable manifold is the Torus $T^n$, since it is covered by $\mathbb{R}^n$, which is clearly hyperspherical. More generally the same holds for any closed manifold admitting a metric of non-positive sectional curvature by the Cartan-Hadamard Theorem. Non-enlargeable manifolds are for example all simply connected manifolds.

The central fact about enlargeable manifolds is that they do not admit psc metrics.

**Theorem 3.2** (GL1980, GL1982, CS2020)**.** *Enlargeable manifolds do not admit metrics of positive scalar curvature.*

This is the result that motivated the original definition by Gromov and Lawson [GL1980], where they first showed that enlargeability is an obstruction to the existence of psc metrics, under the additional assumption that the coverings $\hat{M}$ in Definition 3.1 are finite an spin. They were later able to drop the finiteness assumption [GL1982]. Both proofs are based on techniques using the spin Dirac operator, so it was unclear how and if these results carry over to the non-spin case. Rather recently however, Cecchini and Schick resolved this issue in [CS2020] by relying on modern minimal surface methods developed by Schoen and Yau [SY...].

The following Lemma now provides two easy ways to construct new enlargeable manifolds out of given ones.

**Lemma 3.3.** *(i) The product of enlargeable manifolds is enlargeable.*

*(ii) A closed oriented manifold admitting a non-zero degree map onto an enlargeable manifold is enlargeable.*

*Proof.* For **??** let $M^m$ and $N^n$ be enlargeable. For $\varepsilon > 0$ choose Riemannian coverings $p\colon \hat{M} \to M$ and $q\colon \hat{N} \to N$ as well as $\varepsilon$-contracting maps $f\colon \hat{M} \to S^m$ and $g\colon \hat{N} \to S^n$ of non-zero degree. Then

$$p \times q\colon \hat{M} \times \hat{N} \to M \times N$$

is a Riemannian covering, where both manifolds are equipped with the corresponding product metric. Moreover

$$f \times g\colon \hat{M} \times \hat{N} \to S^m \times S^n$$

defines an $\varepsilon$-contracting map of degree $\deg(f \times g) = \deg(f)\deg(g) \neq 0$. Now consider the map

$$\phi\colon S^m \times S^n \to S^m \wedge S^n \cong S^{m+n},$$

given by projection onto the smash product

$$S^m \wedge S^n := {S^m \times S^n}/{(S^m \times * \cup * \times S^n)},$$

which is well known to be homeomorphic to the sphere $S^{m+n}$. The map $\phi$ is easily seen to be of degree $\pm 1$ (depending on the chosen orientations) and after a slight perturbation we may also assume it to be smooth. Moreover, because $S^m \times S^n$ is closed, it follows that $\phi$ is a $\lambda$-Lipschitz map for some constant $\lambda > 0$. Therefore $\phi \circ (f \times g)$ is a $\lambda\varepsilon$-Lipschitz map of non-zero degree and since $\varepsilon$ was chosen arbitrarily the statement follows.

For **??** let $M^m$ be enlargeable and let $h\colon P \to M$ be some non-zero degree map. Again given $\varepsilon > 0$ choose a Riemannian covering $p\colon \hat{M} \to M$ as well as an $\varepsilon$-contracting map $f\colon \hat{M} \to S^m$. Define a covering $\pi\colon \hat{P} \to P$ as the pullback covering of $p$ under $h$, that is $\pi$ fits into a pullback diagram of the form

$$\begin{tikzcd} \hat{P} \arrow[r, "H"] \arrow[d, "\pi"] & \hat{M} \arrow[d, "p"] \\ P \arrow[r, "h"] & M \end{tikzcd}$$

where $H$ is some suitable smooth map. Since $\hat{M}$ and $P$ are closed, the same holds for $\hat{P}$. Moreover $h$ is $\mu$-Lipschitz for some constant $\mu > 0$, because $P$ is closed. Consequently $H$ is also $\mu$-Lipschitz, given that both $\pi$ and $p$ are local isometries. For the degree of $H$ we compute

$$\deg(p)\deg(H) = \deg(p \circ H) = \deg(h \circ \pi) = \deg(h)\deg(\pi) \neq 0,$$

using the fact that $\deg(h) \neq 0$ by assumption and that the degree of a covering is given by the cardinality of the fiber. We conclude that $H$ is of non-zero degree, such that $f \circ H\colon \hat{P} \to S^m$ is a $\mu\varepsilon$-Lipschitz map of non-zero degree. Again since $\varepsilon$ was chosen arbitrarily the statement follows. □

From Part **??** of the above lemma it follows immediately, that enlargeability is a homotopy invariant, since a homotopy equivalence of connected closed oriented manifolds has degree $\pm 1$. In fact much more is true, namely that enlargeability is a homological invariant in the following sense.

**Theorem** ([**?**]). *Let $M$ be a closed oriented $n$-manifold and let $\phi\colon M \to B\pi_1(M)$ be the classifying map of the universal cover. Then $M$ is enlargeable if and only if $\phi_*[M] \in H_n(B\pi_1(M);\mathbb{Q})$ does not lie in the rational vector subspace*

$$H_n^{\mathrm{small}}(B\pi_1(M);\mathbb{Q}) \subseteq H_n(B\pi_1(M);\mathbb{Q})$$

*of "small" classes, i.e. $\phi_*[M]$ is "large".*

In [**?**] the authors showed this by giving a general definition of "small" and "large" homology classes of a simplicial complex $X$ with finitely generated fundamental group, such that the assignment $X \mapsto H_n^{\mathrm{small}}(X;\mathbb{Q})$ is functorial for continuous maps $f\colon X \to Y$ inducing an epimorphism on $\pi_1$, that is $f_*\colon H_n(X;\mathbb{Q}) \to H_n(Y;\mathbb{Q})$ maps small classes to small classes. Moreover if $f$ happen to induce an isomorphism on $\pi_1$ then $f_*$ also maps large classes to large classes. This general definition then specializes to the fact that a closed oriented manifold $M$ is enlargeable if and only if the fundamental class $[M]$ is large, that is $[M] \notin H_n^{\mathrm{small}}(M;\mathbb{Q})$. Combining these observations, the theorem follows. Furthermore this argument shows that the notion of enlargeability can be purely thought of as a concept of simplicial complexes. For details we refer the reader to [**?**].

## 4 Construction of the Circle Bundles

We will now construct counterexamples to the question raised in the beginning, that is circle bundles $E \to M$ over enlargeable $M$, where the total space $E$ admits a metric of positive scalar curvature.

For $n \geq 2$ let $N^{2n}$ be any $2n$-dimensional totally non-spin manifold. We could for example take $N := \mathbb{C}P^2 \times T^{2n-4}$, which is totally non-spin by Lemma 2.8 and Lemma 2.10. Now consider the manifold

$$M := \mathbb{C}P^n \# T^{2n} \# N.$$

Since the collapse map $M \to T^{2n}$ has degree 1 and $T^{2n}$ is enlargeable, it follows from Lemma 3.3 that $M$ is also enlargeable. We will now construct a circle bundle $E \to M$ as follows. Let

$$c_1 \in H^2(M;\mathbb{Z}) \cong H^2(\mathbb{C}P^n;\mathbb{Z}) \oplus H^2(T^{2n};\mathbb{Z}) \oplus H^2(N;\mathbb{Z})$$

be the cohomology class corresponding to the first chern class $\alpha \in H^2(\mathbb{C}P^n;\mathbb{Z})$ of the canonical bundle $S^{2n+1} \to \mathbb{C}P^n$ under the above isomorphism. We then define $E \to M$ as the unique $S^1$-principal bundle with first chern class equal to $c_1$. Now this is quite an abstract definition, so let us develop an intuition for how this bundle looks like. We claim that $E$ is trivial over the summand $T^{2n}\#N$ and is given by the restriction of the canonical bundle $S^{2n+1} \to \mathbb{C}P^n$ to $\mathbb{C}P^n - D^{2n} \subseteq M$ over the $\mathbb{C}P^n$ summand. To make this precise let

$$i_1 \colon (T^{2n}\#N) - D^{2n} \to M$$

be the inclusion. Then

$$\begin{tikzcd} H^2(M;\mathbb{Z}) \arrow[r, "i_1^*"] & H^2((T^{2n}\#N) - D^{2n};\mathbb{Z}) \\ H^2(\mathbb{C}P^n;\mathbb{Z}) \oplus H^2(T^{2n}\#N;\mathbb{Z}) \arrow[u, "\cong"] \arrow[r, "\text{proj.}"] & H^2(T^{2n}\#N;\mathbb{Z}) \arrow[u, "\cong"] \end{tikzcd}$$

commutes and therefore $i_1^*(c_1) = 0$. Again using the fact that $S^1$-principal bundles are uniquely determined by their first chern class we see that $i_1^*E$ is indeed trivial. Similiarly if we denote by

$$i_2 \colon \mathbb{C}P^n - D^{2n} \to M, \quad j \colon \mathbb{C}P^n - D^{2n} \to \mathbb{C}P^n$$

the inclusions, the diagram

$$\begin{tikzcd} H^2(M;\mathbb{Z}) \arrow[r, "i_2^*"] & H^2(\mathbb{C}P^n - D^{2n};\mathbb{Z}) \\ H^2(\mathbb{C}P^n;\mathbb{Z}) \oplus H^2(T^{2n}\#N;\mathbb{Z}) \arrow[u, "\cong"] \arrow[r, "\text{proj.}"] & H^2(\mathbb{C}P^n;\mathbb{Z}) \arrow[u, "\cong"', "j^*"] \end{tikzcd}$$

commutes and thus $i_2^*(c_1) = j^*(\alpha)$. So the first chern classes of $i_2^*E$ and $j^*S^{2n+1}$ agree, hence they must be isomorphic bundles.

With this intuition in mind we will now show the following two claims.

**Claim 1.** $E$ is totally non-spin.

**Claim 2.** The projection $p \colon E \to M$ induces an isomorphism on $\pi_1$.

Granting these claims for now, let us see how to conclude the existence of a positive scalar curvature metric on $E$. The disk bundle $DE \to M$ corresponding to $E$ defines an oriented null-cobordism of $E$ (see Example 1.6 iii)). Moreover, since $DE \to M$ is a homotopy equivalence it follows from Claim 1 that the inclusion $E \to DE$ is an isomorphism on $\pi_1$. So together with Claim 2 Theorem 2.6 applies, hence $E$ admits a psc metric.

Let us now proof Claim 1 and Claim 2.

*Proof of Claim 1.* Let $i \colon N - D^{2n} \to M$ be the inclusion. We have already established above, that $E$ is trivial over the $T^{2n}\#N$ summand and so in particular

$$i^*E \cong (N - D^{2n}) \times S^1.$$

From Lemma 2.8 (i) and (iii) we conclude that $(N - D^{2n}) \times S^1$ is totally non-spin. Therefore $E$ contains a totally non-spin submanifold of codimension 0 and so Lemma 2.8 (ii) implies that $E$ is also totally non-spin. □

*Proof of Claim 2.* We have to show that the fibers of $E$ can be contracted. This is essentially due to the fact that $E$ contains (the restriction of) the canonical bundle $S^{2n+1} \to \mathbb{C}P^n$ for which the fibers are contractible. To formalize this let $j\colon \mathbb{C}P^n - D^{2n}$ be the inclusion. Consider the following commutative diagram, where the rows are taken from the long exact sequences of homotopy groups of the respective fibrations and the vertical maps are induced by inclusions:

$$\begin{tikzcd}
\pi_2(M) \arrow[r, "\partial_1"] & \pi_1(S^1) \arrow[r] & \pi_1(E) \\
\pi_2(\mathbb{C}P^n - D^{2n}) \arrow[u] \arrow[r, "\partial_2"] \arrow[d, "\cong"'] & \pi_1(S^1) \arrow[u, "\mathrm{id}"'] \arrow[r] \arrow[d, "\mathrm{id}"'] & \pi_1(j^*E) \arrow[u] \arrow[d] \\
\pi_2(\mathbb{C}P^n) \arrow[r, "\partial_3"] & \pi_1(S^1) \arrow[r] & \pi_1(S^{2n+1})
\end{tikzcd}$$

The vertical map on the lower left is an isomorphism, because both $\mathbb{C}P^n$ and $\mathbb{C}P^n - D^{2n}$ are simply connected and so by the Hurewicz Theorem

$$\begin{tikzcd}
\pi_2(\mathbb{C}P^n - D^{2n}) \arrow[r] \arrow[d, "\cong"] & \pi_2(\mathbb{C}P^n) \arrow[d, "\cong"] \\
H_2(\mathbb{C}P^n - D^{2n};\mathbb{Z}) \arrow[r] & H_2(\mathbb{C}P^n;\mathbb{Z})
\end{tikzcd}$$

commutes and the vertical maps are isomorphisms. Since the bottom horizontal map is an isomorphism the same holds for the top horizontal map. We now claim that $\partial_1$ is surjective. For this it suffices to show surjectivity of $\partial_2$, which in turn is equivalent to surjectivity of $\partial_3$. But this is clear by exactness of the rows and $\pi_1(S^{2n+1}) = 0$. Consequently

$$\pi_1(S^1) \to \pi_1(E)$$

is the zero map and therefore we conclude from exactness of

$$\pi_1(S^1) \to \pi_1(E) \to \pi_1(M) \to \pi_0(S^1),$$

that $\pi_1(E) \to \pi_1(M)$ is indeed an isomorphism. □

The above construction now gives examples for circle bundles with positive scalar curvature over enlargeable manifolds $M$ in all even dimensions $\dim(M) \geq 4$. To also get examples in odd dimensions let $E \to M$ be as above and let $\pi\colon M \times S^1 \to M$ be the projection. Then the pullback bundle $\pi^*E \cong E \times S^1$ clearly admits a metric of positive scalar curvature and by Lemma 3.3 the base space $M \times S^1$ is still enlargeable. Naturally, the question arises as to what happens in dimensions $\dim(M) \leq 3$. In this case it turns out that no circle bundle $E \to M$ admits a metric of positive scalar curvature, see [**?**, Theorem B].